

Performance Testing

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Performance Testing:

The objective of performance testing is to evaluate the efficiency, responsiveness, stability, and reliability of the Voice-Based Concept Understanding Analyser (VBCUA) under different workloads. The testing verifies that speech transcription, semantic similarity analysis, audio feature extraction, and report generation execute within acceptable response times while maintaining accuracy.

Step 1: Testing Overview:

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing, Stress Testing
Target Module / API	Flask Prediction API (/predict)
Test Environment	Local System (Windows 10, Python 3.11, Flask)
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	Concurrent Users	Duration (sec)	Expected Outcome
1	Single user uploads and analyzes an audio file	1	30	Successful transcription and evaluation
2	Multiple users upload audio simultaneously	10	60	All requests processed successfully
3	Continuous speech analysis requests	20	120	Stable performance without failures

4	PDF report generation	10	60	Reports generated successfully
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Average)	< 5 sec	0.68 sec	Pass	Fast audio analysis
2	Response Time (Maximum)	< 12 sec	8.2 sec	Pass	Within acceptable limit
3	Speech Transcription Accuracy	> 90 %	95%	Pass	Accurate transcription using Whisper
4	Semantic Evaluation Accuracy	>85%	91%	Pass	Reliable concept evaluation
5	CPU Utilization	< 80%	56%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	62%	Pass	Stable memory

Step 4: Observations & Analysis

Key Findings:

- The VBCUA application successfully transcribed speech using **OpenAI Whisper** with high accuracy.
- Semantic similarity evaluation using **Sentence-BERT** generated reliable concept understanding scores.

Bottlenecks Identified:

- Processing time increases slightly for longer audio recordings (greater than five minutes).
- Initial loading of AI models (Whisper and Sentence-BERT) requires additional startup time.

Optimization Steps Taken:

- Cached AI models to avoid repeated loading.

- Optimized audio preprocessing before transcription.
- Reduced unnecessary intermediate computations during semantic analysis..

Step 5: Screenshots / Evidence

Attached screenshots of the following performance testing outputs:

